

Chen Guan Hsuan

✉ jemmy9211@gmail.com

[in /guan-hsuan-chen](#)
[/jemmy9211](#)
[jemmy9211.github.io](#)

Brief self-introduction

I am a Firmware Engineer at MediaTek with one year of industry experience, specializing in Linux and Zephyr driver development and embedded system development. My current work focuses on developing and testing firmware drivers, verifying FPGA and physical chip functionality, and assisting with chip bring-up activities. I also have experience applying AI technologies to software development, including developing a RAG system and MCP based on an internal kernel driver. With a background in Computer Science and experience in both software engineering and system-level development, I am interested in building efficient and reliable software systems and applying emerging technologies to solve practical problems.

Education

National Tsing Hua University

M.S. Computer Science

September 2023 – July 2025

Hsinchu, Taiwan

National Chung Hsing University

B.S Computer Science

September 2019 – July 2022

Taichung, Taiwan

Work Experience

Mediatek

Firmware Engineer

September 2025 – Now

Hsinchu, Taiwan

- › Develop and test Linux/Zephyr drivers
- › Verify wearable device FPGAs/physical chips
- › Assist with chip bring-up related tasks.
- › Develop a RAG system and MCP based on an internal kernel driver.

Silicon Motion

Firmware Engineer Intern

July 2024 – September 2024

Hsinchu, Taiwan

- › Develop and maintain FTL simulator script builder
- › Learning UFS protocol
- › Assist to create PTL error handling scripts

ASML

Software Engineer Intern

July 2022 – September 2022

Tainan, Taiwan

- › Using python to write testing and automation program helping RD to find Software bug
- › Assist software automation group develop BDD(Behavior-Driven Development) environment
- › Learn skills like Sourcetree(git), Testcomplete, VDI, Python

Research Experience

Software Engineering Laboratory

lab member

September 2023 – July 2025

Hsinchu, Taiwan

- › Department of Computer Science, National Tsing Hua University
- › Research topics such as software testing, code clone detection, commit message generation
- › Set up and maintain the laboratory's deep learning server

Publications

Applying Structured Code Diff Representation and Sequential Hybrid Retrieval to Enhance LLM-Based Commit Message Generation

2025

- › Taiwan Conference on Software Engineering (TCSE 2025)
- › Conference website: <https://tcse2025.seat.org.tw/>

Other Experience

National Tsing Hua University

Teaching Assistant

January 2024 – July 2024

Hsinchu, Taiwan

- › software engineering courses at National Tsing Hua University Department of Computer Science in 2024.
- › Main task is designing assignments and assisting professors.
- › Learn skills like Turnitin, Unit testing, Integration testing, CI/CD, Scrum

Projects

gitbro

- › gitbro is a powerful AI-assisted Git CLI tool that enhances developer productivity by automating commit workflows, summarizing diffs, generating branch names, validating messages, and much more.
- › It supports multiple AI providers including OpenAI, Google Gemini, Anthropic Claude, and locally-hosted Ollama models, allowing both cloud and offline usage. The tool offers an easy-to-use CLI with interactive features, semantic commit support, and built-in pre-commit hook integration for teams.
- › <https://github.com/jemmy9211/gitbro>

TWeather

- › TWeather is a web application that integrates with the public API provided by the Central Weather Bureau of the Republic of China (Taiwan). Utilizing the data offered by this API, the application employs advanced visualization techniques to enhance the accessibility of complex weather data for users. The project leverages Vue, Bootstrap, and Leaflet technologies, facilitating the rapid and practical development of the web interface.
- › **Project Link:** <https://jemmy9211.github.io/TWeather/>

Course

Relevant Courses from September 2018 to July 2024

- * Algorithm (A)
- * Advanced Compiler (A)
- * Computer Organization (A)
- * Formal Language (A-)
- * Computer Programming (A+)
- * Object-oriented Programming (A+)
- * Digital Image Processing (A)
- * Software Reliability and Testing (A+)
- * Linux Kernel (A+)
- * Data Science (A)
- * Parallel Programming (A-)

Club

NCHU Tennis Team

Tfics Programming Club

Skills

Programming Language C/C++, Python, JavaScript

Framework Vue, Django, Flask

Sport Tennis, Swimming

Talents Violin